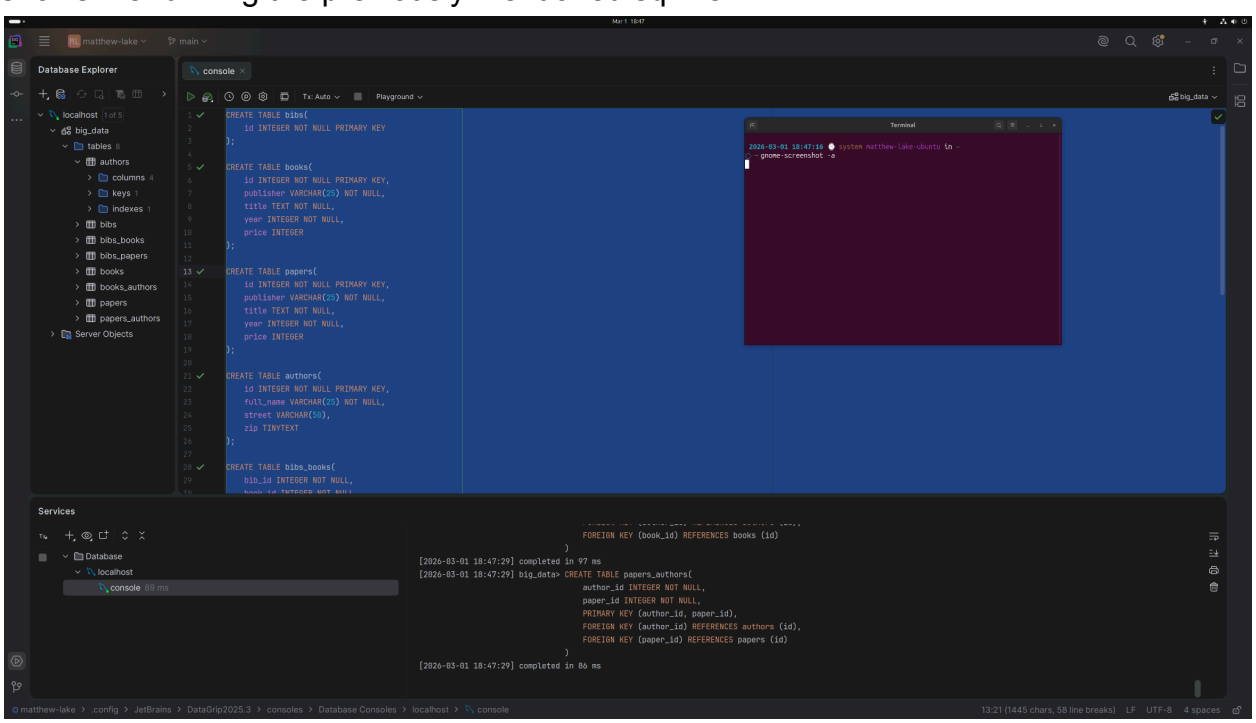
# Lab Assignment 2
# CIS 612
# Matthew Lake
# 2716928

## Introduction

I split this lab up into two parts. I made separate Python scripts for each part. For the first part I extracted the bibs data from the XML input file, created Python data classes for each entity, finally made CSV files from the Python data classes I made. In the second part of the lab I read from those CSV files I made in part one and inserted the data into mySQL database tables.

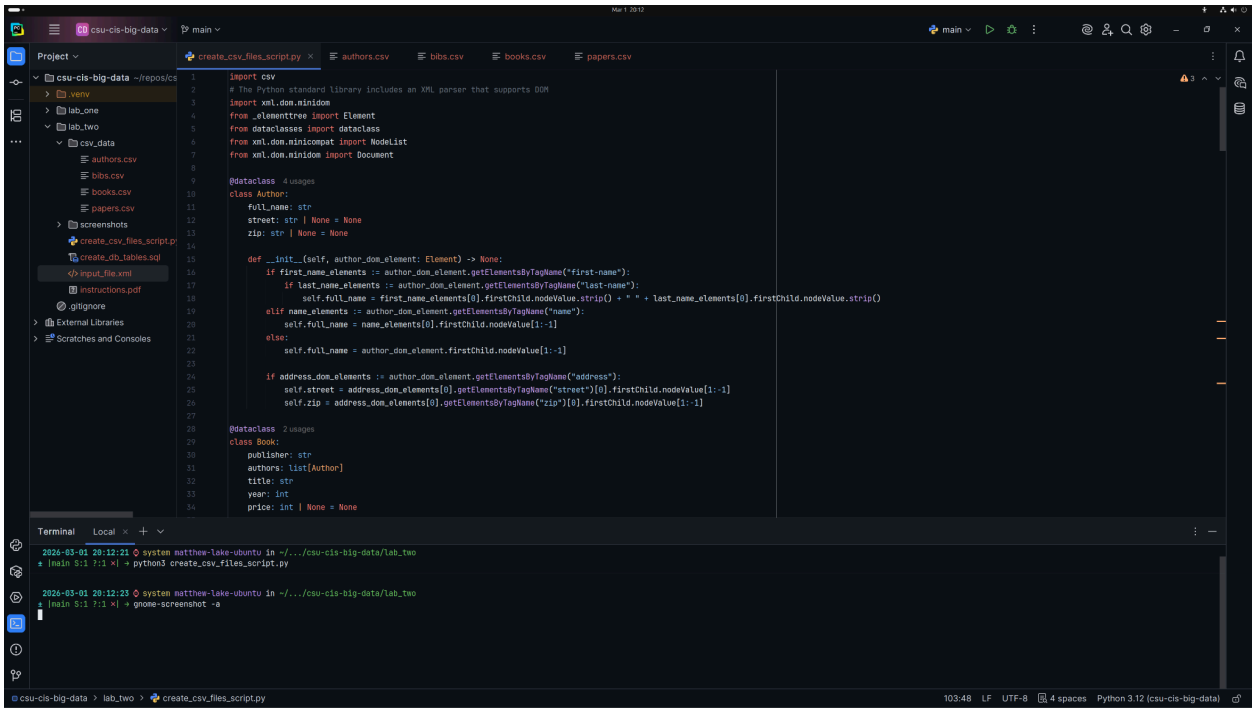
Before I got started creating the scripts I created a database schema based on the input XML file. The table declarations can be found in the `create_db_tables.sql` file. I created eight tables in total. The table names are `bibs`, `books`, `papers`, `bibs_books`, `bibs_papers`, `authors`, `books_authors`, and `papers_authors`. There are four join tables to account for the nested entity relationships in the XML file. The tables `bibs`, `books`, `papers`, and `authors` have an int `id` as their tables primary key. In the join tables the primary keys are the related entity's `id` values. For example, the `book_authors` table has a primary key of `(book_id, author_id)`. The screenshot below shows me running the previously mentioned sql file.

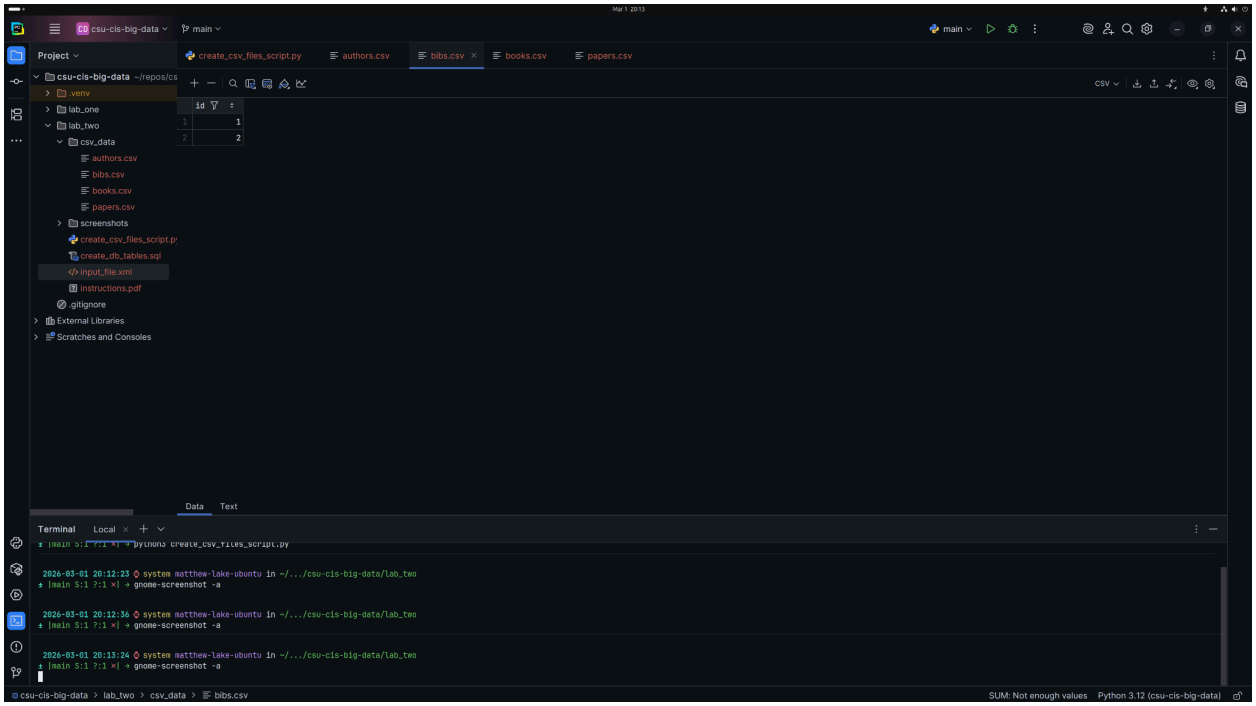


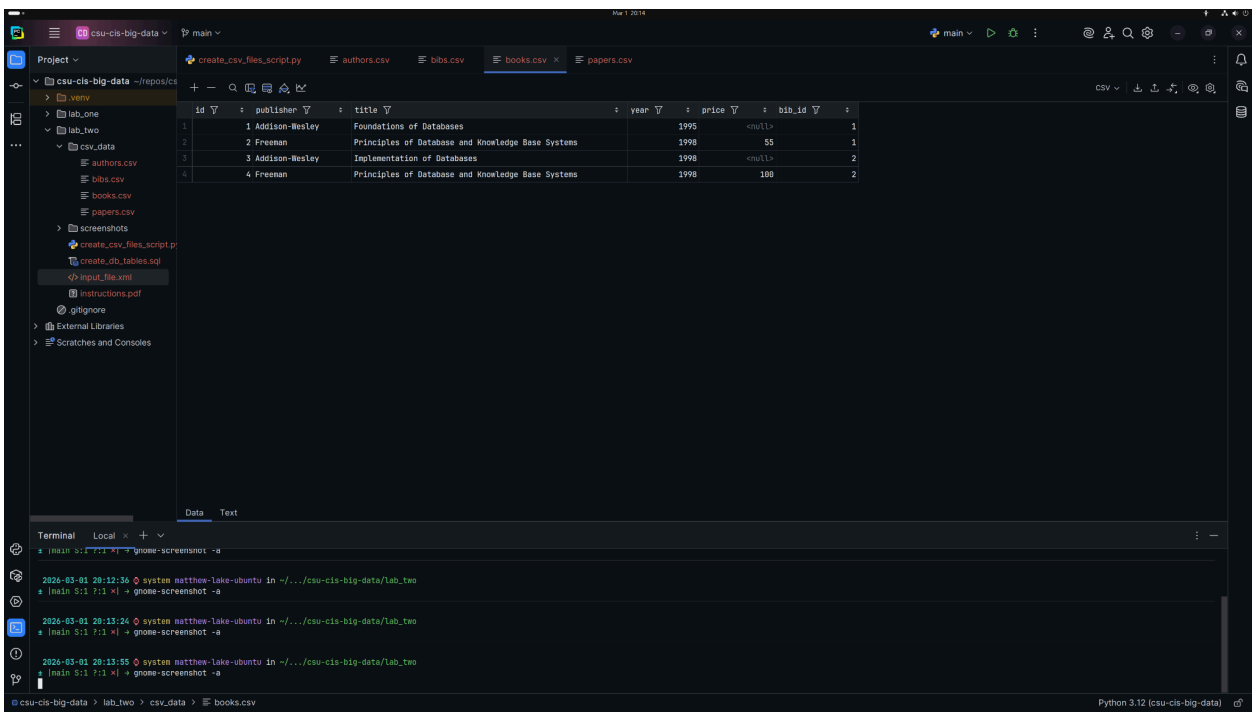
## Part One

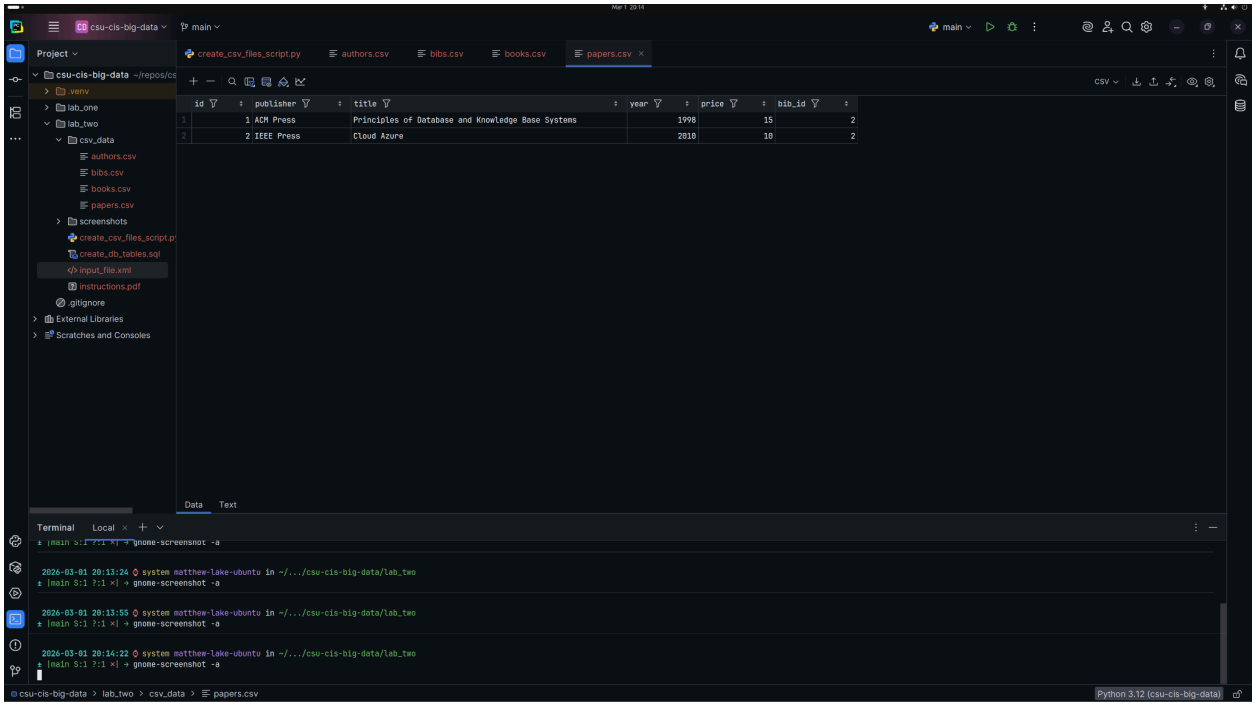
For part one, I created a Python script to first extract the bibs in the XML file and their nested entities books, papers, authors, and address. I utilized the Python native standard library XML parser `xml.dom.minidom`. I combined authors and their address since an author would only have one location. Additionally, I normalized the author names to just consist of one full name. Next, I created dataclasses for all of the entities and their relationships. Finally, I took those objects and created CSV files. I created CSV files for bibs, books, papers, and authors. The script file is called `create_csv_files_script.py`. The CSV files are in the `csv_data` directory. Below are screenshots that show me running the script and showing off the CSV files.

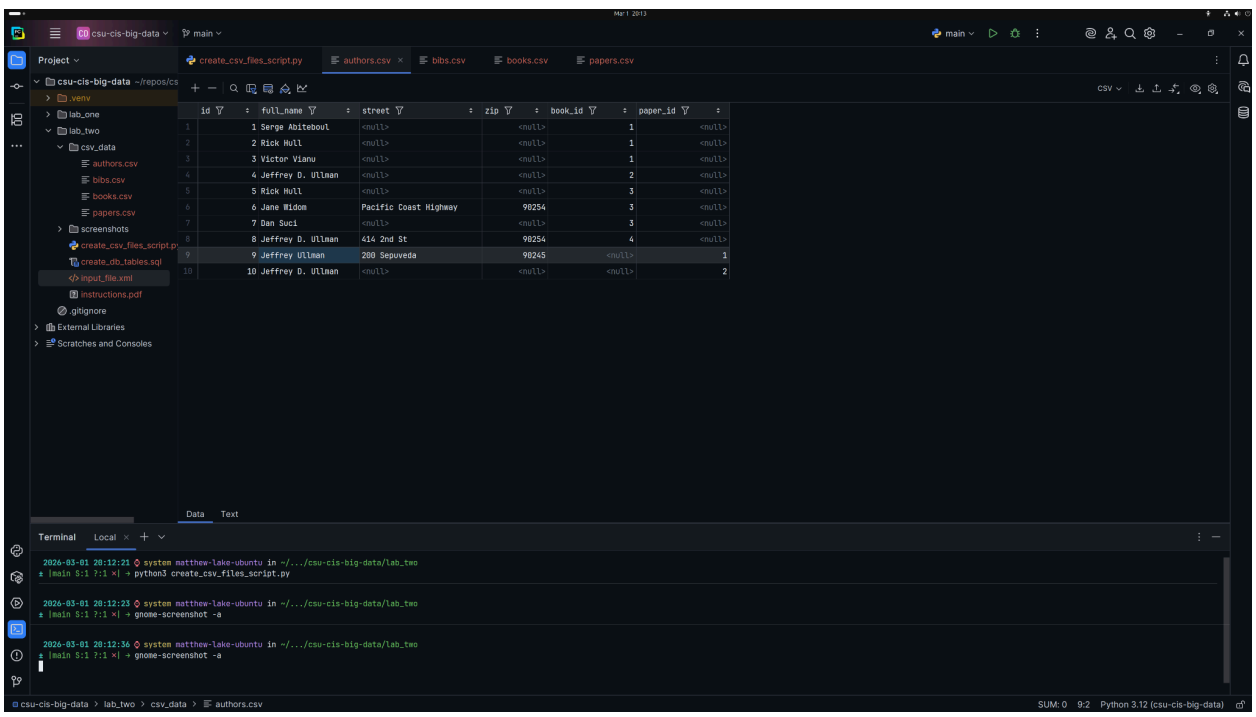
**Note**: Looking back now I should have let the DB determine my id values with `AUTO_INCREMENT`. I handle the ids manually in the scripts and they are stored in the CSV files.




Mar 1 20:13
csu-cis-big-data
main
Project
create_csv_files_script.py
authors.csv
bibs.csv
books.csv
papers.csv
csu-cis-big-data ~/repos/cs
.venv
lab_one
lab_two
csv_data
authors.csv
bibs.csv
books.csv
papers.csv
screenshots
create_csv_files_script.p
create_db_tables.sql
input_file.xml
instructions.pdf
.gitignore
External Libraries
Scratches and Consoles
id
1
2
Data
Text
Terminal
Local
2026-03-01 20:12:23 system matthew-lake-ubuntu in ~/.../csu-cis-big-data/lab_two
± |main S:1 ?:1 ×| → gnome-screenshot -a
2026-03-01 20:12:36 system matthew-lake-ubuntu in ~/.../csu-cis-big-data/lab_two
± |main S:1 ?:1 ×| → gnome-screenshot -a
2026-03-01 20:13:24 system matthew-lake-ubuntu in ~/.../csu-cis-big-data/lab_two
± |main S:1 ?:1 ×| → gnome-screenshot -a
csu-cis-big-data > lab_two > csv_data > bibs.csv
SUM: Not enough values
Python 3.12 (csu-cis-big-data)


Mar 1 20:14
csu-cis-big-data
main
Project
create_csv_files_script.py
authors.csv
bibs.csv
books.csv
papers.csv
csu-cis-big-data ~/repos/cs
.venv
lab_one
lab_two
csv_data
authors.csv
bibs.csv
books.csv
papers.csv
screenshots
create_csv_files_script.p
create_db_tables.sql
input_file.xml
instructions.pdf
.gitignore
External Libraries
Scratches and Consoles
id publisher title year price bib_id
1 Addison-Wesley Foundations of Databases 1995 <null> 1
2 Freeman Principles of Database and Knowledge Base Systems 1998 55 1
3 Addison-Wesley Implementation of Databases 1998 <null> 2
4 Freeman Principles of Database and Knowledge Base Systems 1998 100 2
Data
Text
Terminal
Local
2026-03-01 20:12:36 system matthew-lake-ubuntu in ~/.../csu-cis-big-data/lab_two
± |main S:1 ?:1 ×| → gnome-screenshot -a
2026-03-01 20:13:24 system matthew-lake-ubuntu in ~/.../csu-cis-big-data/lab_two
± |main S:1 ?:1 ×| → gnome-screenshot -a
2026-03-01 20:13:55 system matthew-lake-ubuntu in ~/.../csu-cis-big-data/lab_two
± |main S:1 ?:1 ×| → gnome-screenshot -a
csu-cis-big-data > lab_two > csv_data > books.csv
Python 3.12 (csu-cis-big-data)

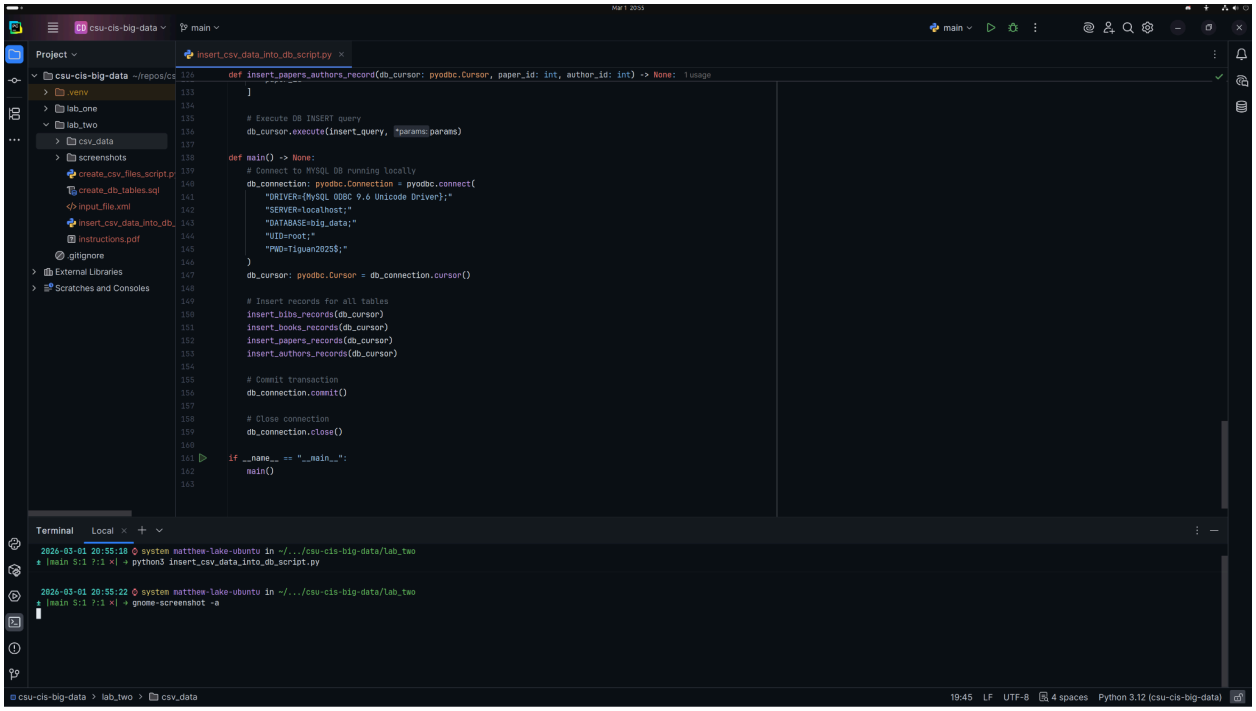


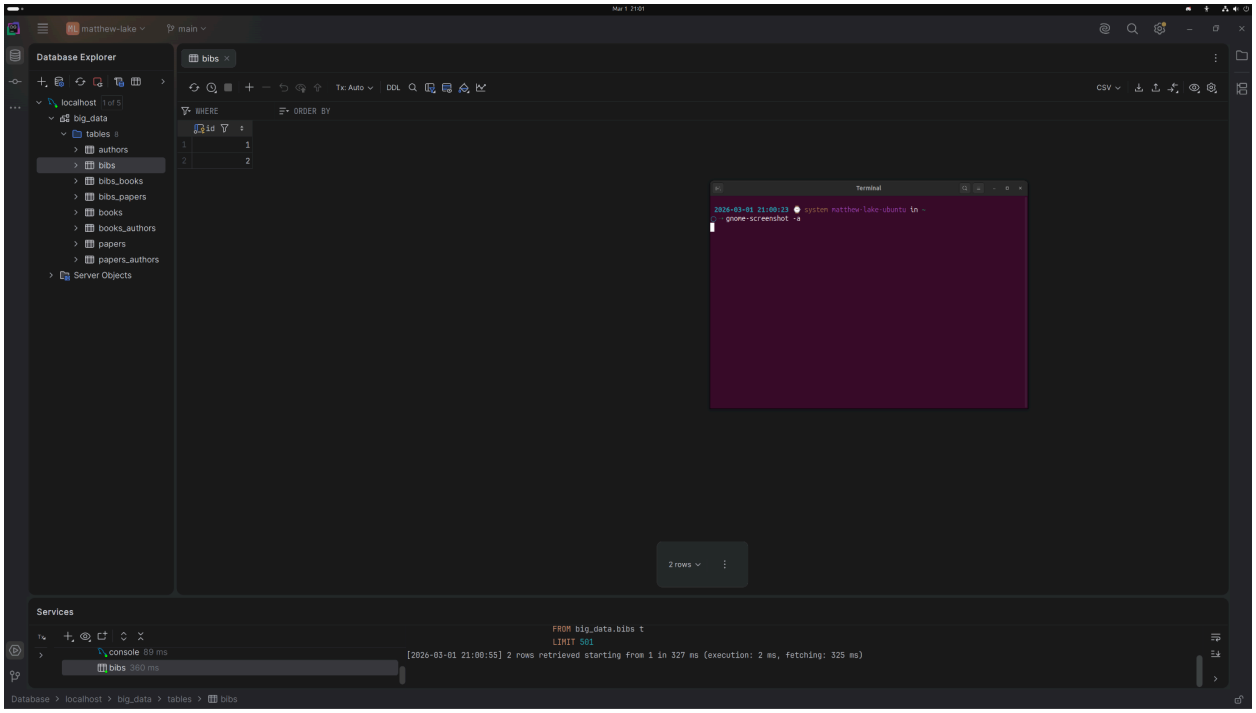



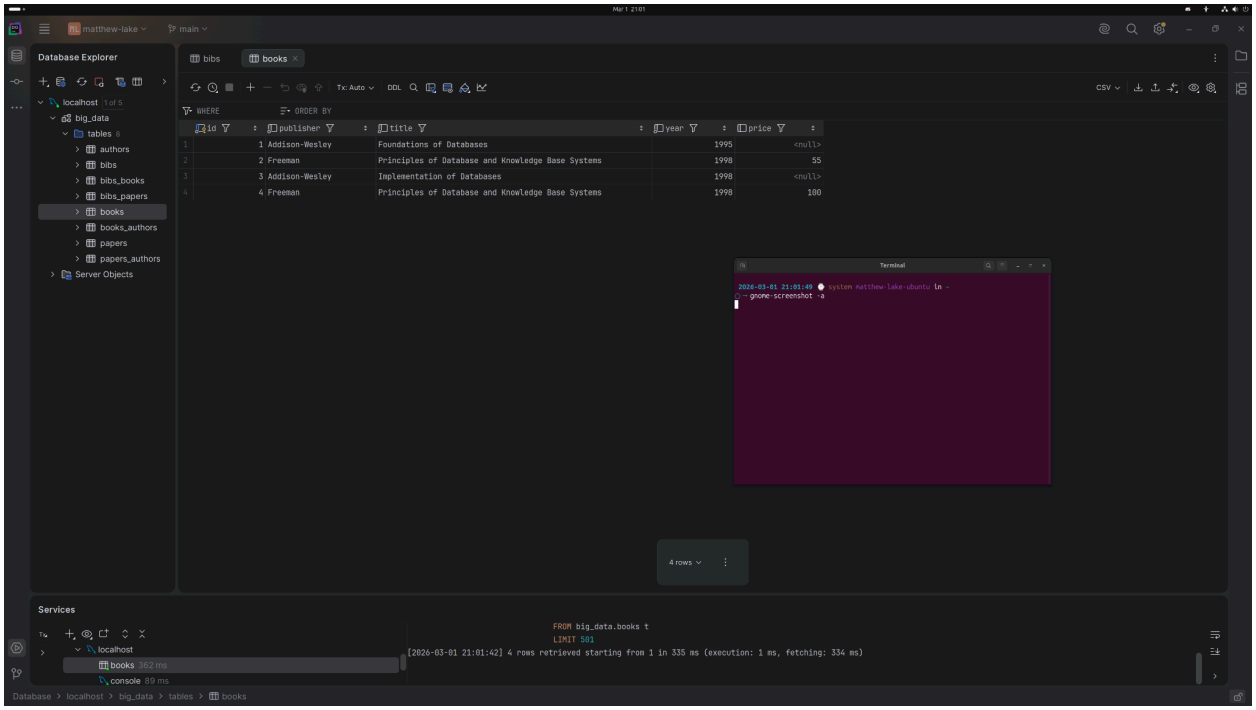

# Part 2

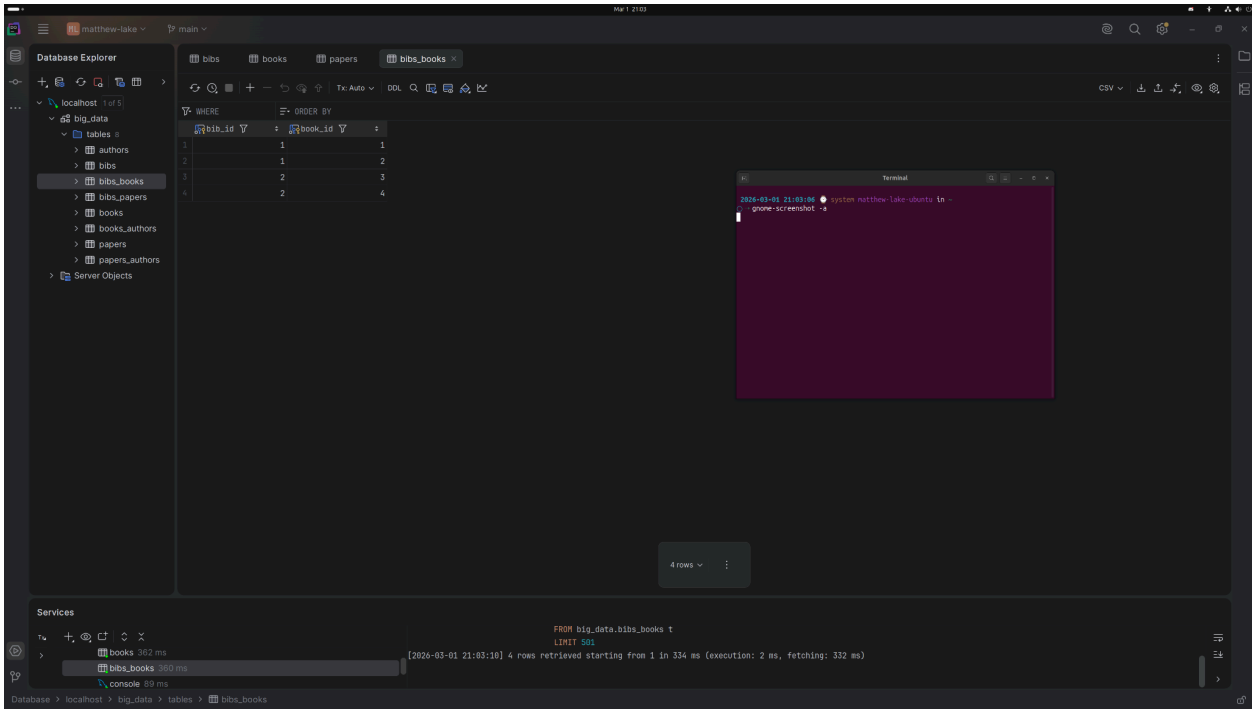
For part two I extracted the data from the CSV files made in Part one and inserted the resulting records into their respective tables. The Python script file is called `insert_csv_data_into_db_script`. Below are screenshots showing me running my script

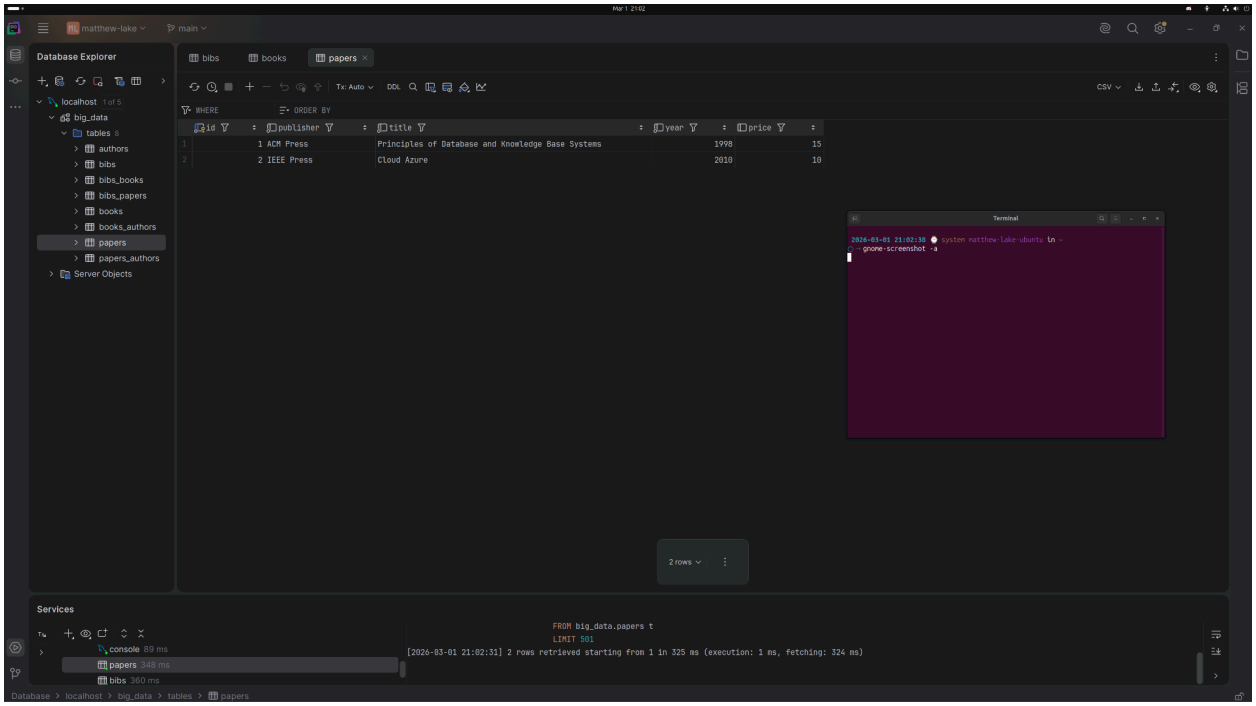
and what the tables look like after all of the data are in them.

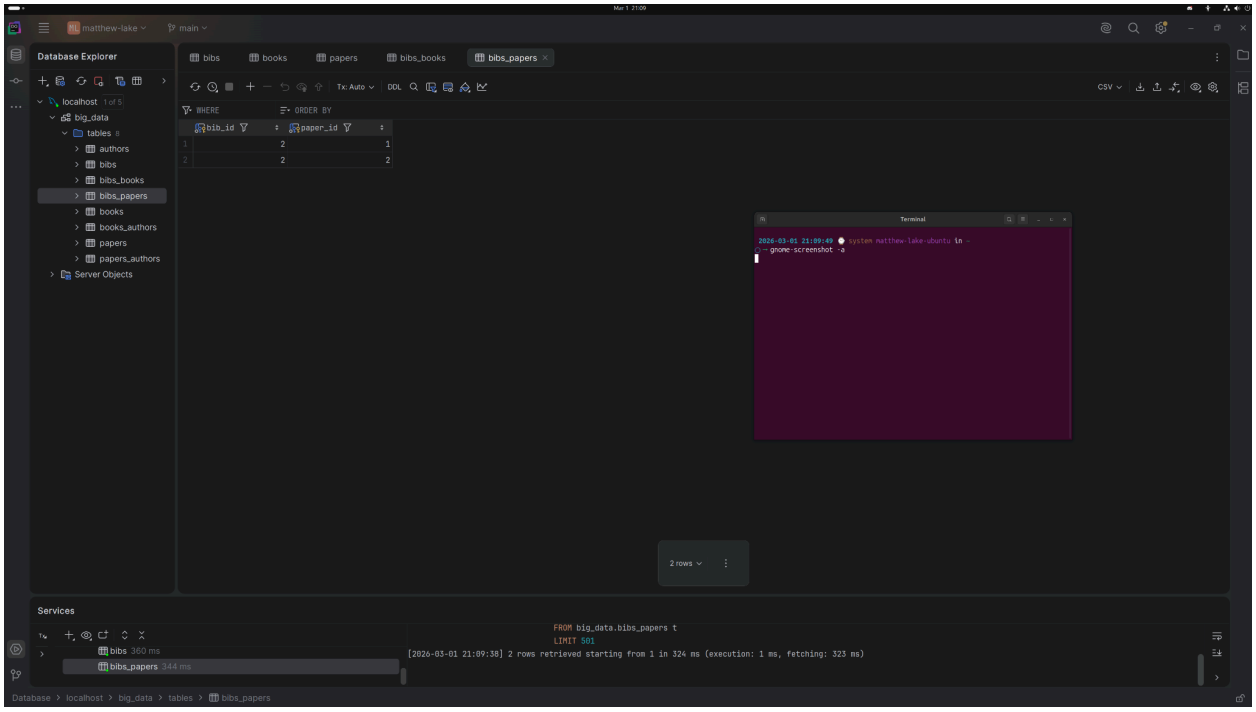


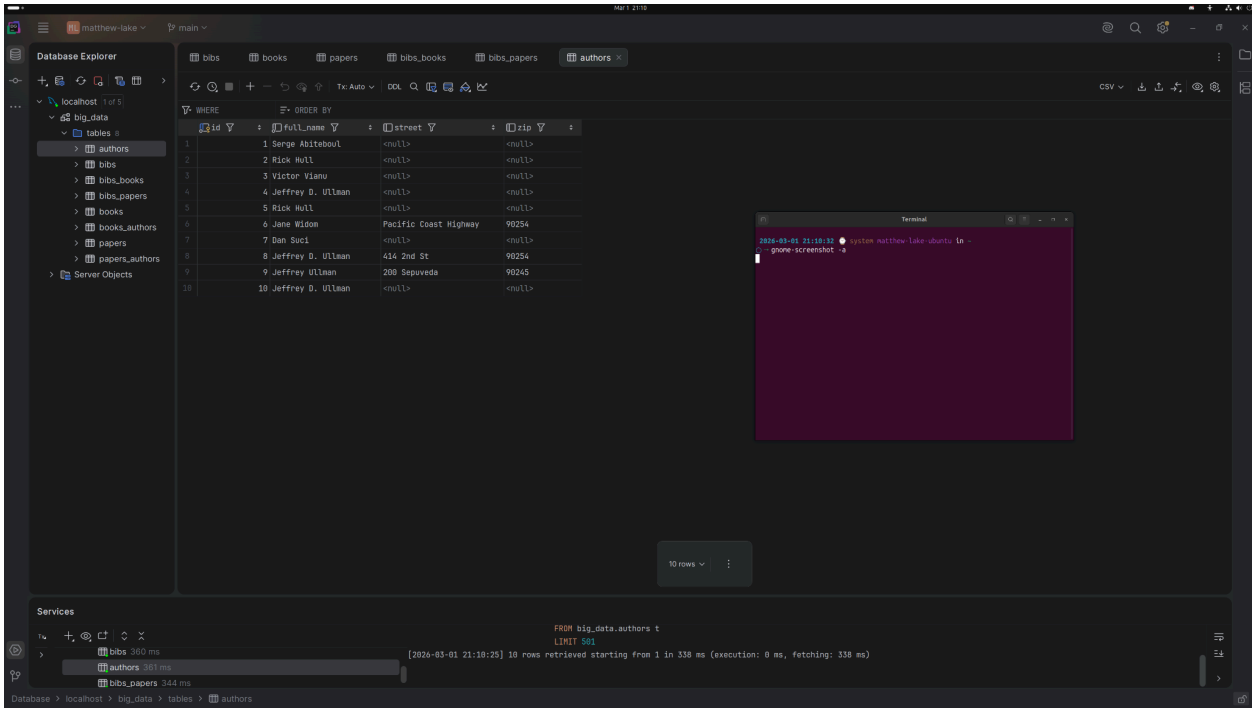



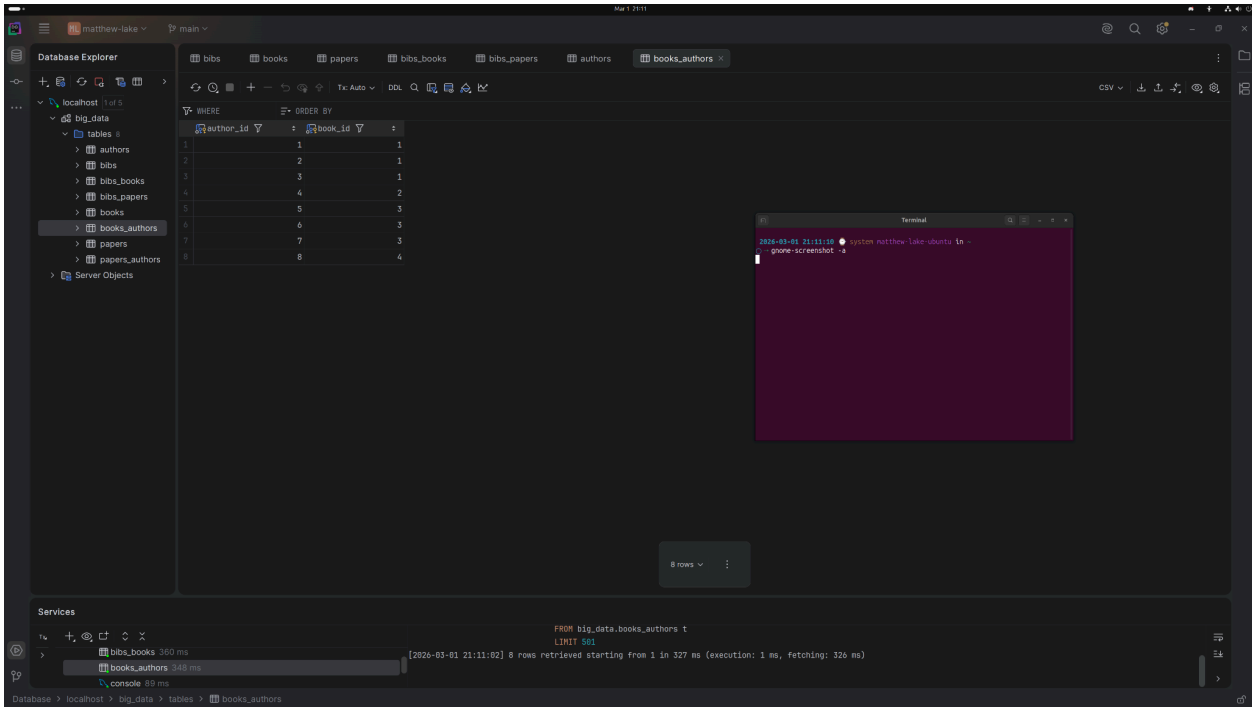

Mar 1 21:01
matthew-lake
main
Database Explorer
bibs
books
Tx: Auto
DDL
CSV
WHERE
ORDER BY
localhost 1 of 5
big_data
tables 8
authors
bibs
bibs_books
bibs_papers
books
books_authors
papers
papers_authors
Server Objects
id publisher title year price
1 Addison-Wesley Foundations of Databases 1995 <null>
2 Freeman Principles of Database and Knowledge Base Systems 1998 55
3 Addison-Wesley Implementation of Databases 1998 <null>
4 Freeman Principles of Database and Knowledge Base Systems 1998 100
Terminal
2026-03-01 21:01:49 system matthew-lake-ubuntu in ~
gnome-screenshot -a
4 rows
Services
FROM big_data.books t
LIMIT 501
[2026-03-01 21:01:42] 4 rows retrieved starting from 1 in 335 ms (execution: 1 ms, fetching: 334 ms)
localhost
books 362 ms
console 89 ms
Database > localhost > big_data > tables > books

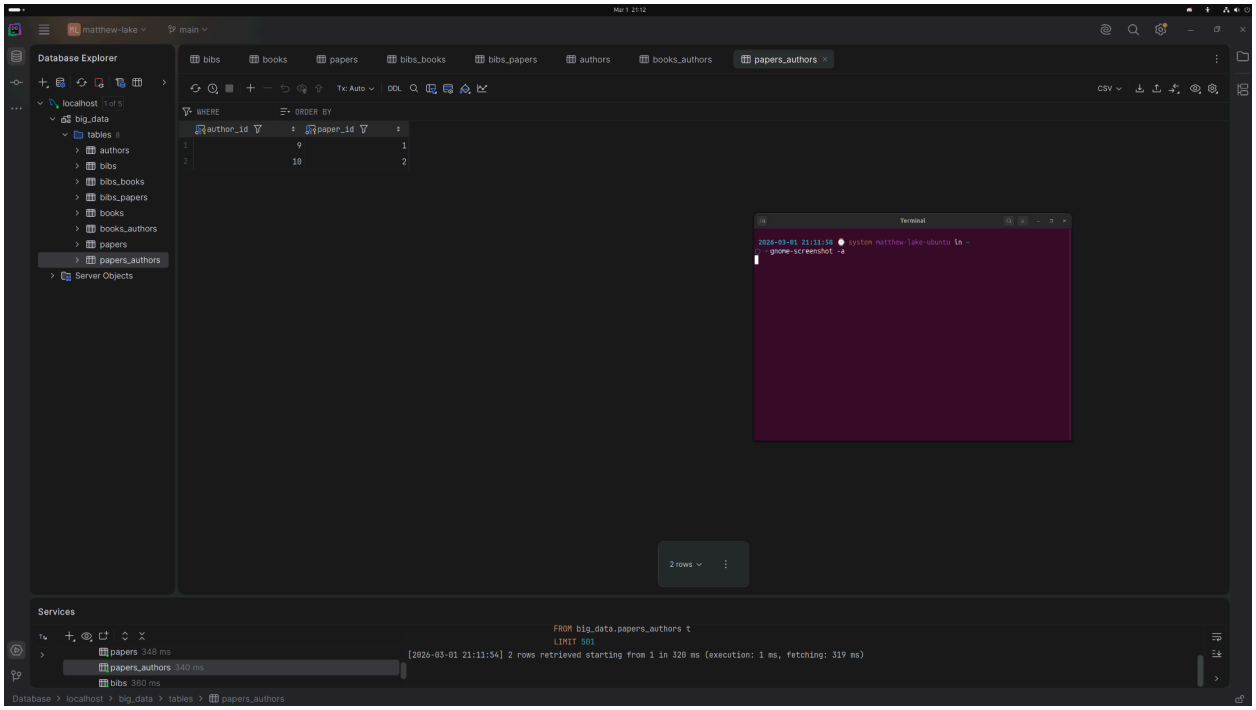

Mar 1 21:03
matthew-lake
main
Database Explorer
bibs
books
papers
bibs_books
Tx: Auto
DDL
CSV
WHERE
ORDER BY
localhost 1 of 5
big_data
tables 8
authors
bibs
bibs_books
bibs_papers
books
books_authors
papers
papers_authors
Server Objects
bib_id book_id
1 1
1 2
2 3
2 4
Terminal
2026-03-01 21:03:06 system matthew-lake-ubuntu in ~
gnome-screenshot -a
4 rows
Services
FROM big_data.bibs_books t
LIMIT 501
[2026-03-01 21:03:10] 4 rows retrieved starting from 1 in 334 ms (execution: 2 ms, fetching: 332 ms)
books 362 ms
bibs_books 360 ms
console 89 ms
Database > localhost > big_data > tables > bibs_books


Database Explorer
bibs
books
papers
WHERE
ORDER BY
id | publisher | title | year | price
1 | ACM Press | Principles of Database and Knowledge Base Systems | 1998 | 15
2 | IEEE Press | Cloud Azure | 2010 | 10
Terminal
2026-03-01 21:02:38 system matthew-lake-ubuntu in ~
gnome-screenshot -a
2 rows
Services
FROM big_data.papers t
LIMIT 501
[2026-03-01 21:02:31] 2 rows retrieved starting from 1 in 325 ms (execution: 1 ms, fetching: 324 ms)
Database > localhost > big_data > tables > papers


Database Explorer
bibs
books
papers
bibs_books
bibs_papers
WHERE
ORDER BY
bib_id | paper_id
2 | 1
2 | 2
Terminal
2026-03-01 21:09:49 system matthew-lake-ubuntu in ~
gnome-screenshot -a
2 rows
Services
FROM big_data.bibs_papers t
LIMIT 501
[2026-03-01 21:09:38] 2 rows retrieved starting from 1 in 324 ms (execution: 1 ms, fetching: 323 ms)
Database > localhost > big_data > tables > bibs_papers


Database Explorer
bibs
books
papers
bibs_books
bibs_papers
authors
id
full_name
street
zip
1 Serge Abiteboul
2 Rick Hull
3 Victor Vianu
4 Jeffrey D. Ullman
5 Rick Hull
6 Jane Widom Pacific Coast Highway 90254
7 Dan Suci
8 Jeffrey D. Ullman 414 2nd St 90254
9 Jeffrey Ullman 200 Sepuveda 90245
10 Jeffrey D. Ullman
10 rows
Terminal
gnome-screenshot -a
Services
FROM big_data.authors t
LIMIT 501
[2026-03-01 21:10:25] 10 rows retrieved starting from 1 in 338 ms (execution: 0 ms, fetching: 338 ms)


Database Explorer
bibs
books
papers
bibs_books
bibs_papers
authors
books_authors
author_id
book_id
8 rows
Terminal
gnome-screenshot -a
Services
FROM big_data.books_authors t
LIMIT 501
[2026-03-01 21:11:02] 8 rows retrieved starting from 1 in 327 ms (execution: 1 ms, fetching: 326 ms)


Mar 1 21:12
matthew-lake
main
Database Explorer
bibs
books
papers
bibs_books
bibs_papers
authors
books_authors
papers_authors
Tx: Auto
DDL
CSV
WHERE
ORDER BY
localhost 1 of 5
big_data
tables 8
authors
bibs
bibs_books
bibs_papers
books
books_authors
papers
papers_authors
Server Objects
author_id
paper_id
9
1
10
2
Terminal
2026-03-01 21:11:52 system matthew-lake-ubuntu ~
gnome-screenshot -a
2 rows
Services
FROM big_data.papers_authors t
LIMIT 501
papers 348 ms
papers_authors 340 ms
bibs 360 ms
[2026-03-01 21:11:54] 2 rows retrieved starting from 1 in 320 ms (execution: 1 ms, fetching: 319 ms)
Database > localhost > big_data > tables > papers_authors